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A Software Approach to the PPT2 Conjecture

MASTER'S THESIS

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Intellectual Property Statement

Statement text regarding intellectual property goes here...

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Abstract

This thesis explores...

Povzetek

V tem delu raziskujemo...

Razširjeni povzetek

Daljši slovenski povzetek vsebine...

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TODO

- ☐ write introduction - what is entanglement, how do we use it, what is the PPT2 conjecture and why it matters, what are its implications if true or false. Current state of knowledge i.e. known results for $n \leq 3$ expectations for $n \geq 4$, testing entanglement is hard...
- ☐ generating PPT candidates, 1. start with psd matrices AA^T (optionally symmetrize) and add smallest eigenvalue of partial transpose on diagonal to ensure PPT - the downside here is that the state may not be entangled, 2. use unextendible product bases (UPB) - this should guarantee bound entanglement, i.e. PPT and entangled, requires references to UPB papers and some explanation
- ☐ A note on composition calculation; not as straightforward as multiplying matrices, needs the Choi-Jamiołkowski transformation so that we can work with matrices, define *ampliation* $(I \otimes C_\Phi)(C_\Psi)$ which we use to compute the Choi matrix of the composition, as well check entanglement
- ☐ DPS-based entanglement detection, describe hierarchy, SDPs in general, witness extraction for comparison later, extensions of DPS, i.e. adding KKT constraints, our code uses existing implementations (cite Ket library)
- ☐ Abhisheks PnCP map generation method, describe in more detail what it does and how it works, Julia implementation does rationalization differently for performance reasons, i.e. only after computing SDP, explain why this still works, get matlab and julia performance numbers to compare, explain how the matrix representation is chosen for each map (not unique), chosen to be invariant to partial transpose, but doesn't need to be - discussion
- ☐ get references for all statements, known results, methods, etc.
- ☐ write out entire algorithm/pipeline - loop: 1. generate candidate(s), 2. compute composition, 3. test for entanglement (using DPS and/or precomputed witnesses), 4. if candidate is entangled, rationalize and retest to confirm, 5. if confirmed, export certificate and stop
- ☐ results: no counterexamples found on 10000 random candidates, in 4x4 this takes about five hours to verify using DPS level 2 and a collection of 10000 precomputed witnesses
- ☐ issues with generated entanglement witnesses: almost surely not optimal, possibly even decomposable, so probably fails to detect most entangled states
- ☐ issues with DPS: numerical instability, only lower levels of hierarchy are computationally feasible
- ☐ required sections: 1. introduction - explains entanglement, PPT2 conjecture 2. theoretical background - required formalism: positivity, complete positivity, entanglement, partial transpose, quantum states/channels, entanglement witnesses, DPS hierarchy, semidefinite programming, Choi representation, maps/matrices/operators 3. methods - generating candidates, why we can't use known constructions, compositions, testing entanglement via DPS and witnesses, generating witnesses via PNCP maps, post-solver numerical validation, performance notes 4. results/discussion/conclusion - TBD

careful with “improvements” on the DPS hierarchy; some of them work only for specific types of states i.e. diagonal unitary invariant states, for which the PPT² conjecture has already been proven

Introduction

Quantum entanglement is a central nonclassical resource in quantum information theory. It supports communication and cryptographic tasks and provides a structural lens for understanding quantum channels [1].

This thesis studies the PPT² conjecture: whether the composition of two PPT maps is always entanglement breaking [2]. The conjecture has a clear operational interpretation, since entanglement-breaking maps destroy all bipartite entanglement with any reference system.

Known progress motivates a computation-first strategy. The conjecture is proven in low-dimensional regimes and in specific families such as Choi-type maps [3], [4]. In higher dimensions, the geometry of positive maps becomes more complex, and direct analytic classification remains difficult.

The core challenge is twofold. First, separability testing is hard in the regimes where the conjecture remains open. Second, numerical certificates near feasibility boundaries are fragile. Therefore, this thesis focuses on methodology: how to construct, compose, test, and validate candidate objects in a reproducible way.

The main objectives are:

1. Design a Julia workflow for candidate generation, map composition, and entanglement checks.
2. Integrate DPS-based semidefinite relaxations as a baseline witness route.
3. Integrate an SOS-based PNCP witness-construction route with post-solver validation.

The rest of the thesis is organized as follows. Chapter 2 provides mathematical prerequisites and formulations. Chapter 3 presents the computational workflow and implementation architecture. Chapter 4 discusses methodological limitations and numerical reliability. Chapter 5 concludes with implications and next directions.

Theoretical Background

This chapter establishes the mathematical framework underpinning the PPT² conjecture and its computational study. We fix notation and define the key objects — linear maps, their positivity properties, quantum states, and entanglement criteria — then develop the semidefinite programming tools used in the implementation.

Preliminaries

Notation and matrix spaces

We write M_n for the algebra of complex $n \times n$ matrices, $H_n \subseteq M_n$ for the subspace of Hermitian matrices ($A = A^*$), and $M_n^+ \subseteq H_n$ for the cone of positive semidefinite matrices.

Definition 1 (*Positive semidefinite and positive definite*) A matrix $A \in H_n$ is *positive semidefinite* (PSD), written $A \succeq 0$, if $x^*Ax \geq 0$ for all $x \in \mathbb{C}^n$, or equivalently if all eigenvalues of A are non-negative. If all eigenvalues are strictly positive, A is *positive definite* (PD), written $A \succ 0$.

The *Hilbert-Schmidt inner product* on H_n is

$$\langle A, B \rangle := \text{Tr}[AB], \quad (1)$$

making H_n an inner product space. For general M_n the inner product is $\langle A, B \rangle = \text{Tr}[A^*B]$.

We work with bipartite systems on the tensor product space $\mathbb{C}^m \otimes \mathbb{C}^n$ (subsystem dimensions m and n). The operator space is $M_m \otimes M_n \simeq M_{mn}$. We write $E_{ij} \in M_n$ for the matrix unit with a 1 in position (i, j) and zeros elsewhere.

Linear maps on matrix spaces

Definition 2 (*Positive, k -positive, and completely positive maps*) Let $\Phi : M_n \rightarrow M_m$ be a linear map.

- Φ is *positive* (P) if $A \succeq 0 \Rightarrow \Phi(A) \succeq 0$.
- Φ is *k -positive* if $I_k \otimes \Phi : M_k \otimes M_n \rightarrow M_k \otimes M_m$ is positive.
- Φ is *completely positive* (CP) if it is k -positive for every $k \in \mathbb{N}$ [3].

Every CP map is positive. Maps that are positive but not completely positive (PNCP) play a central role: they detect entanglement in a way that CP maps cannot.

Definition 3 (*Choi-Jamiołkowski isomorphism*) The *Choi matrix* of a linear map $\Phi : M_n \rightarrow M_m$ is

$$C_\Phi = \sum_{i,j=1}^n E_{ij} \otimes \Phi(E_{ij}) \in M_n \otimes M_m. \quad (2)$$

The assignment $\Phi \mapsto C_\Phi$ is a linear isomorphism between maps $\Phi : M_n \rightarrow M_m$ and matrices in $M_n \otimes M_m$. The map is recovered by $\Phi(A) = \text{Tr}_1[(A^T \otimes I_m)C_\Phi]$.

Under this isomorphism, Φ is CP if and only if $C_\Phi \succeq 0$ [5], and Φ is trace-preserving if and only if $\text{Tr}_B[C_\Phi] = I_n$.

Definition 4 (*Partial transpose*) For a product matrix $A \otimes B \in M_m \otimes M_n$, the *partial transpose* with respect to subsystem B is defined as

$$(A \otimes B)^{\Gamma_B} := A \otimes B^T. \quad (3)$$

Extending by linearity to a general $\rho \in M_m \otimes M_n$, written in block form as $\rho = (\rho_{ij})_{i,j=1}^m$ with blocks $\rho_{ij} \in M_n$:

$$\rho^\Gamma := (I_m \otimes T)(\rho) = \left(\rho_{ij}^T\right)_{i,j=1}^m, \quad (4)$$

where T is the transposition map. For Hermitian ρ , the partial transpose with respect to either subsystem yields the same spectrum: $(\rho^{\Gamma_A})^T = \rho^{\Gamma_B}$, so $\rho^{\Gamma_A} \geq 0 \iff \rho^{\Gamma_B} \geq 0$. We therefore write ρ^Γ without specifying the subsystem.

Definition 5 (*PPT state and PPT map*) A state $\rho \in M_m \otimes M_n$ is *PPT* if $\rho \geq 0$ and $\rho^\Gamma \geq 0$. A map $\Phi : M_n \rightarrow M_m$ is *PPT* if its Choi matrix is PPT:

$$C_\Phi \geq 0 \quad \text{and} \quad C_\Phi^\Gamma \geq 0. \quad (5)$$

Since $C_\Phi \geq 0$ characterizes CP maps, a PPT map is a CP map whose Choi matrix has non-negative partial transpose. The composition of two PPT maps is trivially PPT.

Map composition via Choi matrices

The Choi matrix of the composition $\Phi \circ \Psi$ is not the product $C_\Phi C_\Psi$. Instead, it is obtained by applying the *ampliation* $I_n \otimes \Phi$ to C_Ψ :

$$C_{\Phi \circ \Psi} = (I_n \otimes \Phi)(C_\Psi), \quad (6)$$

where the right-hand side applies Φ block-wise to each $n \times n$ block of C_Ψ . Expanding in the standard basis yields the index formula

$$(C_{\Phi \circ \Psi})_{ip,jq} = \sum_{k,l} (C_\Psi)_{ik,jl} (C_\Phi)_{kp,lq}. \quad (7)$$

This formula is used directly in the implementation to compose two candidate PPT maps from their Choi matrices.

PPT2 Conjecture

With PPT maps and entanglement-breaking maps (Definition Definition 2 below) in hand, we can now state the central object of study.

Theorem 1 (*PPT2 Conjecture*) If Φ_1 and Φ_2 are PPT maps, then $\Phi_1 \circ \Phi_2$ is entanglement breaking [2].

The conjecture is proven for $n = 2$ (since all PPT states in $M_2 \otimes M_n$ are separable [6]), for $n = 3$ [3], and for Choi-type maps in all dimensions [4]. It remains open for $n \geq 4$ in the general case. The rest of this chapter develops the formalism needed to study the conjecture computationally.

Quantum States and Entanglement

A *quantum state* on $\mathbb{C}^n$ is a density matrix: a PSD matrix $\rho \in M_n^+$ with $\text{Tr}[\rho] = 1$. A *quantum channel* is a completely positive trace-preserving (CPTP) map $\Phi : M_n \rightarrow M_m$.

Definition 1 (*Separability and entanglement*) A bipartite state $\rho \in M_m \otimes M_n$ is *separable* if

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B, \quad p_i \geq 0, \quad \sum_i p_i = 1, \quad \rho_i^A \in M_m^+, \quad \rho_i^B \in M_n^+. \quad (8)$$

Otherwise ρ is *entangled*.

Deciding separability is NP-hard in general [7]. Since an exact test is computationally intractable, one relies on necessary conditions: a state that fails such a condition must be entangled.

Definition 2 (*Entanglement-breaking map*) A map $\Phi : M_n \rightarrow M_m$ is *entanglement breaking* (EB) if $(I_k \otimes \Phi)(\rho)$ is separable for every $k \in \mathbb{N}$ and every state $\rho \in M_k \otimes M_n$ [1].

Testing $k = 1$ alone would only check that Φ preserves separability of bipartite states; the ampliation over all k ensures Φ destroys entanglement with any external reference system. Equivalently, Φ is EB if and only if its Choi matrix C_Φ is a separable state [1].

Separability criteria

Since testing separability exactly is NP-hard, one uses necessary conditions. A state that violates such a condition is certifiably entangled; satisfying all known criteria does not guarantee separability. For a comprehensive survey see [8].

PPT criterion. If ρ is separable, then $\rho^\Gamma \geq 0$ [6], [9]. More precisely: the transposition map T is positive but not completely positive, so applying $I \otimes T$ to an entangled state can produce a non-PSD result. If $\rho^\Gamma \not\geq 0$, then ρ is entangled. The condition is necessary but not sufficient: bound entangled (PPT entangled) states exist whenever $mn > 6$ [1].

Range criterion. If $\rho \in M_m \otimes M_n$ is separable, there exist product vectors $\{|\psi_i\rangle \otimes |\varphi_i\rangle\}$ spanning the range of ρ such that $\{|\psi_i\rangle \otimes \overline{|\varphi_i\rangle}\}$ spans the range of ρ^Γ [10]. The range criterion can detect certain PPT entangled states, but fails when ρ is full rank (e.g., under noise), since then any set of vectors spans its range trivially.

Reduction criterion. If $\rho \in M_m \otimes M_n$ is separable, then

$$\rho_A \otimes I_n - \rho \geq 0 \quad \text{and} \quad I_m \otimes \rho_B - \rho \geq 0, \quad (9)$$

where $\rho_A = \text{Tr}_B[\rho]$ and $\rho_B = \text{Tr}_A[\rho]$ are the *reduced states* (obtained by tracing out one subsystem) [11].

CCNR criterion. Define the *realignment* of ρ as the matrix $R(\rho)$ with entries $(R(\rho))_{im+k, jn+l} = \rho_{in+j, kn+l}$. If ρ is separable, then $\|R(\rho)\|_1 \leq 1$, where $\|\cdot\|_1$ is the trace norm [12]. The CCNR criterion is independent of the PPT criterion and can detect some PPT entangled states that the other criteria miss.

Majorization criterion. If ρ is separable, then $\lambda(\rho) \preceq \lambda(\rho_A)$ and $\lambda(\rho) \preceq \lambda(\rho_B)$, where $\lambda(\cdot)$ denotes the non-increasingly ordered eigenvalue vector [13]. Here $a \preceq b$ denotes *majorization*: $\sum_{i=1}^k a_i \leq \sum_{i=1}^k b_i$ for all k , with equality at $k = n$. (Note: $\leq$ here is majorization order, distinct from the positive-definiteness notation $\succeq$.) This criterion follows from the reduction criterion [13] and shares its limitations.

Beyond criterion-based tests, *algorithmic* approaches reformulate separability as a convex optimization problem. The DPS hierarchy (Section Section 3.5) is the most powerful

general-purpose method. Criteria based on covariance matrices and entanglement witnesses (including Bell inequalities) cover complementary cases but are less directly applicable in our setting [8].

Entanglement witnesses

Definition 3 (*Entanglement witness*) A Hermitian operator W is an *entanglement witness* if $\text{Tr}[W\sigma] \geq 0$ for all separable σ , and $\text{Tr}[W\rho] < 0$ for some entangled ρ .

The set of separable states is convex and closed. By the Hahn-Banach separation theorem — any point outside a closed convex set is separated from it by a hyperplane — for every entangled $\rho \notin \text{SEP}$ there exists a Hermitian operator W (a hyperplane in H_{mn}) with $\text{Tr}[W\rho] < 0$ and $\text{Tr}[W\sigma] \geq 0$ for all $\sigma \in \text{SEP}$ [1]. Geometrically, each witness W cuts off a half-space that contains ρ but not SEP ; the intersection of all such half-spaces recovers SEP exactly.

figure: sep-
arable set
and wit-
ness hyper-
plane

Witnesses are one-sided: $\text{Tr}[W\rho] < 0$ certifies entanglement, but $\text{Tr}[W\rho] \geq 0$ does not certify separability.

Definition 4 (*Decomposable witness*) A witness W is *decomposable* if $W = P + Q^\Gamma$ for some $P, Q \geq 0$; otherwise it is *non-decomposable* [14].

Decomposable witnesses cannot detect PPT entangled states: for any PPT σ , $\text{Tr}[(P + Q^\Gamma)\sigma] = \text{Tr}[P\sigma] + \text{Tr}[Q\sigma^\Gamma] \geq 0$. Only non-decomposable witnesses are useful for our purposes.

Under the Choi-Jamiołkowski isomorphism, every entanglement witness $W = C_\Phi$ corresponds to a PNCP map Φ , and vice versa. The map condition $(I_k \otimes \Phi)(\rho) \neq 0$ is strictly stronger than the scalar condition $\text{Tr}[C_\Phi\rho] < 0$ [1].

Semidefinite Programming

A *semidefinite program* (SDP) is a convex optimization problem in which a linear objective is minimized subject to a linear matrix inequality. The standard primal form is

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && F(x) := F_0 + \sum_{i=1}^n x_i F_i \geq 0, \end{aligned} \tag{10}$$

where $x \in \mathbb{R}^n$ is the optimization variable, $c \in \mathbb{R}^n$ is the cost vector, and $F_0, F_1, \dots, F_n \in H_d$ are fixed Hermitian matrices. The associated *dual SDP* is

$$\begin{aligned} & \text{maximize} && -\text{Tr}[F_0 Z] \\ & \text{subject to} && Z \geq 0, \\ & && \text{Tr}[F_i Z] = c_i, \quad i = 1, \dots, n. \end{aligned} \tag{11}$$

The *Slater condition* for Equation 10 requires the existence of a strictly feasible point: some x with $F(x) > 0$. When it holds for both primal and dual, *strong duality* holds: primal and dual optima coincide. When $c = 0$, Equation 10 is a *feasibility problem*. If infeasible, a dual feasible $Z \geq 0$ with $\text{Tr}[F_i Z] = 0$ and $\text{Tr}[F_0 Z] > 0$ certifies infeasibility.

Interior point methods

SDPs are solved in practice by *interior point methods* (IPMs), which follow the *central path* — a smooth trajectory through the strict interior of the feasible region parameterized by a barrier coefficient $\mu \rightarrow 0$. A standard IPM applied to a $d \times d$ SDP with n variables reaches ε -accuracy in $O(\sqrt{d} \log(1/\varepsilon))$ iterations, each requiring $O(n^2 d^2 + nd^3)$ operations [15]. This thesis uses MOSEK [16], a state-of-the-art IPM solver for semidefinite programs.

A property of IPMs important for our use: solutions lie in the *strict interior* of the feasible region ($F(\mathbf{x}) > 0$), which provides numerical slack around the boundary. This is exploited in the post-solver rationalization (Section Section 3.6.4). However, all solutions are floating-point approximations, and any conclusion near a feasibility boundary requires explicit validation.

Entanglement testing as an SDP

Separability testing is NP-hard [7]. One standard formulation maximizes the expectation of an observable M over separable states:

$$h_{\text{SEP}}(M) = \max\{\text{Tr}(M\rho) : \rho \in \text{SEP}\}. \quad (12)$$

The DPS hierarchy (Section Section 3.5) relaxes this problem by replacing SEP with a sequence of increasingly tight SDP-representable outer approximations.

DPS Hierarchy

The Doherty-Parrilo-Spedalieri (DPS) hierarchy [17] provides a sequence of SDP relaxations of separability that is *complete*: for every entangled state there exists a finite level k at which the test detects it. In principle, running the hierarchy to convergence solves separability exactly; in practice, only the first few levels are computationally feasible.

Symmetric extensions

For a separable state $\rho = \sum_i \lambda_i x_i x_i^* \otimes y_i y_i^*$, define the k -fold *symmetric extension*

$$\rho_k = \sum_i \lambda_i x_i x_i^* \otimes (y_i y_i^*)^{\otimes k} \in H(\mathbb{C}^m \otimes (\mathbb{C}^n)^{\otimes k}). \quad (13)$$

This extension satisfies three properties, each with a physical interpretation:

$$(I_m \otimes \Pi_k) \rho_k (I_m \otimes \Pi_k) = \rho_k, \quad (14)$$

$$\rho_k^{\Gamma_s} \geq 0 \quad \forall s = 1, \dots, k, \quad (15)$$

$$\text{Tr}_{B_2 \dots B_k}[\rho_k] = \rho. \quad (16)$$

Here Π_k projects onto the bosonic (fully symmetric) subspace of $(\mathbb{C}^n)^{\otimes k}$. Condition Equation 14 states that ρ_k is invariant under permutations of the k copies of B . Condition Equation 15 states that all partial transposes of ρ_k are PSD. Condition Equation 16 states that tracing out the extra $k - 1$ copies of B recovers ρ .

Definition 1 (*DPS set at level k*) DPS_n^k is the set of states $\rho \in H(\mathbb{C}^m \otimes \mathbb{C}^n)$ for which there exists a symmetric extension $\rho_k \in H(\mathbb{C}^m \otimes (\mathbb{C}^n)^{\otimes k})$ satisfying Equation 14, Equation 15, and Equation 16.

Each DPS_n^k is defined by semidefinite constraints, so membership is testable in polynomial time for fixed k . The hierarchy satisfies [17]:

1. $\text{SEP}_n \subseteq \text{DPS}_n^k$ and $\text{DPS}_n^{k+1} \subseteq \text{DPS}_n^k$ for all $k \geq 1$.
2. DPS_n^1 is equivalent to the PPT criterion.
3. Asymptotic completeness: $\bigcap_{k \geq 1} \text{DPS}_n^k = \text{SEP}_n$.

Feasibility SDP and witness extraction

Testing $\rho \in \text{DPS}_n^k$ amounts to searching for ρ_k satisfying Equation 14, Equation 15, and Equation 16: this is an SDP feasibility problem. When it is infeasible (no valid extension exists), the dual variable $Z \succeq 0$ yields an entanglement witness.

Concretely, the SDP dual to the level- k test has a feasible Z whenever ρ is entangled at that level. The operator

$$W = \text{Tr}_{B_2 \dots B_k}[Z] \tag{17}$$

is an entanglement witness for ρ [17]: one can verify $\text{Tr}[W\rho] < 0$ directly from Z , without relying on the floating-point primal solution. This dual certificate is how we confirm any entanglement detection numerically.

Improvements and limitations

Several enhancements to the basic hierarchy are known. Harrow, Natarajan, and Wu [18] add first-order optimality (KKT) conditions to the DPS SDP, achieving *finite convergence*: infeasibility is certified at a finite level rather than only asymptotically. The KKT conditions are linear constraints on the Lagrange multipliers of the original SDP; they increase the variable count substantially and make clean witness extraction from the dual more involved.

Specialized hierarchies have been developed for states with symmetry. For diagonal unitary invariant states, Britz and Laurent [19] give a drastically smaller SDP at each level. For Werner states and isotropic states, explicit separability conditions are known. However, none of these apply to our search: the PPT2 conjecture is already proven for the relevant symmetric families [4], so our search must use the general DPS hierarchy.

Sum-of-Squares Polynomials and PNCP Maps

A complementary approach to DPS exploits the polynomial representation of linear maps. The key insight: positivity of a map is equivalent to non-negativity of a polynomial, and complete positivity is equivalent to that polynomial being a sum of squares (SOS). Since SOS is characterized by a semidefinite condition (via Gram matrices), this connects map positivity directly to SDPs.

Non-negative polynomials and SOS

Let $\mathbb{R}[x, y]$ be the ring of real polynomials in $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$. A polynomial p is *non-negative* if $p(x, y) \geq 0$ for all real inputs; it is a *sum of squares* (SOS) if $p = \sum_i q_i^2$ for polynomials q_i . Every SOS polynomial is non-negative, but not every non-negative polynomial is SOS — the gap between the two is the source of PNCP maps.

Gram matrix and SDP connection

Definition 1 (*Gram matrix representation*) A homogeneous polynomial p of degree $2d$ is SOS if and only if there exists $G \succeq 0$ such that

$$p(z) = v(z)^T G v(z), \quad (18)$$

where $v(z)$ is the vector of monomials of degree d in z .

Testing whether p is SOS therefore reduces to a semidefinite feasibility problem: find $G \succeq 0$ satisfying the linear constraints that equate coefficients of Equation 18 with those of p . If no such G exists, p is not SOS.

Polynomial representation of maps

Each linear map $\Phi : M_n \rightarrow M_m$ corresponds to a biquadratic polynomial:

$$p_\Phi(x, y) := y^T \Phi(xx^T) y. \quad (19)$$

“Non-negative” here means $p_\Phi(x, y) \geq 0$ for all real (x, y) — the polynomial analogue of positivity for matrices. The fundamental correspondence [20] is:

- Φ is *positive* if and only if p_Φ is non-negative on $\mathbb{R}^n \times \mathbb{R}^m$.
- Φ is *completely positive* if and only if p_Φ is SOS.

A PNCP map corresponds exactly to a non-negative non-SOS polynomial, and constructing such a polynomial gives an entanglement witness directly.

Algorithm 1: KMSZ construction for PNCP maps

- 1: Sample random rational points $x^{(1)}, \dots, x^{(t)} \in \mathbb{Q}^n$ and $y^{(1)}, \dots, y^{(t)} \in \mathbb{Q}^m$.
 - 2: Form rank-1 bilinear forms $h_j(x, y) = \langle x^{(j)}, x \rangle \cdot \langle y^{(j)}, y \rangle$. Each h_j is a product of two linear forms, so $\sum_j h_j^2$ is SOS.
 - 3: Find $f \notin \text{span}\{h_1, \dots, h_t\}$ over $\mathbb{Q}$ such that f is non-negative and not SOS. This uses linear algebra: choose f orthogonal to the SOS cone within the space of degree-2 biquadratic forms [20].
 - 4: Solve the SDP: find the smallest $\delta > 0$ such that $F_\delta := \delta f + \sum_j h_j^2 \geq 0$. For small enough δ , the SOS term $\sum h_j^2$ dominates δf , ensuring non-negativity.
-

Steps 1–3 are over $\mathbb{Q}$ and produce rational h_j, f . The map F_δ is non-negative by construction and non-SOS because f contributes a non-SOS term, so the corresponding map is PNCP [20].

Rationalization

Since step 4 of Algorithm 1 involves a floating-point SDP, δ is only known numerically. To obtain an exact rational certificate, we rationalize the Gram matrix post hoc.

Because F_δ is non-SOS, we cannot directly find a Gram matrix for it. Instead, we relax: search for a Gram matrix G for $F_\delta \cdot S$, where $S = \left(\sum_i x_i^2 + \sum_j y_j^2\right)^l$ is a fixed SOS multiplier. Expanding $F_\delta \cdot S$ in the monomial basis $\mathbf{v}$ produces the feasibility problem

$$G \succeq 0, \quad \langle A_i, G \rangle = b_i, \quad i = 1, \dots, m, \quad (20)$$

with rational data A_i, b_i from the coefficient equations. If Equation 20 is feasible, then $F_\delta \cdot S$ is SOS, which certifies that F_δ is non-negative [20].

Let G be a numerical solution with $\mu = \min \text{eig}(G) > 0$ and residual $\varepsilon = \max_i |\langle A_i, G \rangle - b_i|$. If $\mu > \varepsilon$, a rational feasible $\hat{G}$ is obtained by:

1. Exploiting the known e -dimensional nullspace [20]: set the corresponding blocks $\tilde{G}_{11} = 0$, $\tilde{G}_{12} = 0$.
2. Rationalize the remaining block $\tilde{G}_{22}$ using continued fractions.
3. Project back to the affine constraint space $\langle A_i, G \rangle = b_i$.
4. Verify $\hat{G} \succeq 0$ by a final rational feasibility SDP.

Once $\hat{G}$ is rationalized, the coefficients of F_δ are extracted by subtracting the contribution of the SOS multiplier S from $\mathbf{v}^T \hat{G} \mathbf{v}$. A final SDP check verifies that the extracted polynomial is not SOS, confirming the PNCP certificate is exact. Because IPMs produce solutions in the strict interior, the gap $\mu > \varepsilon$ holds with room to spare, making the rationalization robust in practice.

Methods

This chapter defines the computational workflow used in the current implementation.

Although channels are trace preserving by definition, the computational tests in this thesis are primarily positivity/sign based. Therefore, intermediate computations may use non-normalized maps, then normalize where needed for interpretation.

A quantum state is usually represented by a positive semidefinite operator normalized with unit trace. In this work, we mainly work with unnormalized quantum states and consider its convex cone.

The goal is reproducibility with mathematical traceability: each code step corresponds to a defined object or operation from Chapter 2.

Design constraints:

1. separability testing difficulty in relevant dimensions,
2. numerical fragility near SDP feasibility boundaries,

PPT Maps

Algorithm 2: Generating PPT maps

- 1: Generate a matrix R with entries sampled from a normal distribution.
 - 2: $A = R^T R$ is a positive semidefinite matrix.
-

- 3: Optionally symmetrize each block of A to ensure the resulting map is invariant under partial transpose.
- 4: Adjust the spectrum of A to ensure the resulting map is PPT: $A_{PPT} = A - \lambda I$, where $\lambda < 0$ is the smallest eigenvalue of A^Γ .

Composite maps and entanglement testing

Compute Composite PPT maps from generated candidates and test with DPS and precomputed witnesses

Algorithm 3: Randomized PPT Composition + DPS

- 1: Generate random PPT maps Φ_1, Φ_2 using the method described above.
 - 2: Compute the Choi matrix of the composition C_{Φ_1, Φ_2} using the ampliation operation.
 - 3: Test the resulting state for entanglement using DPS hierarchy and/or precomputed witnesses.
-

Performance

Our performance gains stem from many factors. The MATLAB implementation originally generated random integer matrices to produce nicer results, but the downside is that this approach is not random enough, and often failed. Resulting in a single candidate requiring many recomputation attempts. The Julia implementation allws for such candidate generation, but with speed in mind we opted for the more reliable approach of sampling from a normal distribution, which is far more likely to produce a valid candidate on the first try. While the solver's performance is generally independent of the language, the Julia implementation benefits from more efficient data handling and better integration with the solver, leading to faster overall runtimes. Additionally the rationalization steps in MATLAB are noticeably more expensive, both due to the overhead of rational arithmetic and the choice to rationalize the entire problem beforehand, while in Julia we only rationalize the solution after solving the SDP. All this to say, direct comparison between implementations is somewhat difficult and arguably pointless. It would be like comapring apples to oranges. With that in mind we present some rough performance numbers for the current implementation: The previous implementation was made with other goals in mind, but producing candidates for the same dimensions took orders of magnitude more time:

Add performance numbers

Rationalization and Certificate Validation

Our constructions are intrinsically numerical, and the resulting maps are represented as floating-point matrices. This is sufficient for quick tests and exploration, but numerical noise can lead to false positives near the boundary of positivity. Therefore, we apply a post-solver rationalization and re-validation step to confirm any positive detections.

MATLAB perf computed/from article

The rationalization as explained previously is implemented in the `_fix_gram` function and works as follows:

1. Extract Gram matrix G from constraints of the SDP $(\delta f + h^2)(\sum x_i^2 y_j^2)^l = \mathbf{x}^T \mathbf{y}^T G \mathbf{y} \mathbf{x}$.

2. Compute eigendecomposition of G , set the first e eigenvalues equal to zero, and rationalize the remaining coefficients to obtain $\tilde{G}$.
3. Reformulating the constraints in terms of $\tilde{G}$, isolate the coefficients of polynomial $\hat{p} = \delta f + h^2$ from the relaxation terms $(\sum x_i^2 y_j^2)^l$.
4. Solve another feasibility SDP to verify $\hat{p}$ is not a sum of squares.

Implementation Architecture

The active implementation is in the code directory, centered on `code/src/ppt2.jl`.

details on code architecture - main stuff in `src` folder, additional scripts for reproducibility, experiments notebooks, dependencie - general julia stuff, any JuMP compatible solvers, Ket as existing quantum toolbox, and usage instructions

Complexity and Practical Limits

All the problems we are looking at are SDPs with exponential [7] complexity in dimension and relaxation depth [17]. Practical bottlenecks are:

1. processing power: these problems are by nature not easily parallelizable, simplex methods are intrinsically sequential and internal point methods rely on factoring large *sparse* matrices, so the potential of GPU speedups is fairly low.
2. memory growth: the size of the SDP grows exponentially with dimension and relaxation depth, leading to memory bottlenecks even for moderate dimensions, i.e. for 4×4 states DPS level 3 is already infeasible on standard hardware, requiring hundreds of gigabytes of RAM. Even improvements such as adding KKT constraints to achieve finite convergence [18] lead to significant increases in problem size, so they are not without cost.
3. solver instability near degeneracy: SDP solvers produce floating-point solutions, and when the problem is near the boundary of feasibility, small numerical errors can lead to incorrect conclusions about positivity. This is particularly problematic for our purposes, since we are interested in the very existence of such bound states.
4. Generated entanglement witnesses are not guaranteed to be optimal, there has been research on finding optimal witnesses [14], but this only applies to specific cases, so it remains open, how our witnesses can be improved. In fact many of our witnesses may even be decomposable, which would make them unable to detect PPT-entangled states [1].
5. Searching for PPT candidates in the first place is non-trivial, and random generation may not be sufficient to find counterexamples if they exist. We may require a more structured approach if the volume of the search space is in fact 0 (this would not mean the conjecture holds).

Experimental Evaluation

This section summarizes computational results from the pipeline implementations.

Computational Results

All our tests were performed for the 4×4 case, since it is the smallest dimension where the conjecture remains open. While our methods may handle even slightly higher dimensions, the computational cost grows rapidly, and we have not yet fully explored those regimes.

We have computed a library of 10,000 PNCP maps using the KMSZ construction. Each map was verified to be positive but not completely positive through rationalization and SOS checks.

Using Pipeline A (Randomized PPT Composition + DPS), we performed broad-coverage searches on random PPT-map compositions:

- 10,000 random candidates generated with PPT constraints
- DPS level 2 in addition to the precomputed entanglement witnesses
- Computational time: approximately 5 hours on 80 CPU cores for complete verification

No counterexamples to the PPT2 conjecture were found in these searches.

Conclusion

This thesis presents a computational approach to the PPT2 conjecture, integrating methods for generating PPT maps, composing them, and testing for entanglement using both DPS-based relaxations and PNCP-based witness construction. While no counterexamples were found in our extensive searches, the methodology provides a framework for further exploration and potential discovery in this open problem.

Links

- <https://arxiv.org/pdf/1506.08834>
- <https://arxiv.org/pdf/quant-ph/0603199>
- <https://arxiv.org/pdf/2402.12944>
- <https://arxiv.org/pdf/1807.01266>
- <https://arxiv.org/pdf/2001.01181>
- <https://arxiv.org/pdf/2010.07898>
- <https://arxiv.org/pdf/1807.03636>
- <https://arxiv.org/pdf/2011.03809>
- <https://arxiv.org/pdf/2512.06551>
- <https://arxiv.org/html/2501.03959v1>
- <https://arxiv.org/pdf/quant-ph/0308032>
- <https://felixleditzky.info/teaching/ST23/Felix%20Leditzky%20-%20Math%20595%20Quantum%20channels.pdf>
- <https://arxiv.org/pdf/quant-ph/0602223>
- <https://arxiv.org/pdf/0810.4507>
- <https://arxiv.org/pdf/1309.7992>
- <https://journals.aps.org/prabstract/10.1103/PhysRevA.71.032333>
- <https://journals.aps.org/prresearch/abstract/10.1103/PhysRevResearch.3.023101>

- <https://arxiv.org/pdf/quant-ph/9801069>
- <https://arxiv.org/pdf/quant-ph/0702225>
- <https://arxiv.org/pdf/0907.4979>
- <https://link.springer.com/article/10.1007/s00220-017-2859-0>
- http://congres.cran.univ-lorraine.fr/2002/CDC_2002/pdffiles/papers/832_FrP06-6.pdf
- <https://arxiv.org/html/2506.11346v3>
- <https://arxiv.org/html/2308.07019v5#bib.bib22>
- <https://helda.helsinki.fi/server/api/core/bitstreams/486c1a74-82f8-4c6b-b665-6a9142606c78/content>
- <https://arxiv.org/pdf/0712.1114>
- <https://www.nature.com/articles/s41598-022-14920-5>
- <https://arxiv.org/pdf/0907.2369>
- <https://iopscience.iop.org/article/10.1088/0305-4470/39/45/020/pdf>
- <https://arxiv.org/pdf/0805.1318>
- <https://iopscience.iop.org/article/10.1088/1751-8121/acaa16>
- <https://link.springer.com/article/10.1007/s00023-023-01325-x>

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